

1 1:

In this section we will begin by outlining some of the common notation with further notation introduced in the literature review. ...

At the core of statistical inference is the challenge of mathematically capturing the structure and uncertainty inherent in observed data. ...

Suppose that we have a statistical model $P_\theta \in \mathcal{P}$, where $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$, with the parameters $\theta \in \Theta$ for the data points $x_{1:n} = \{x_1, x_2, \dots, x_n\}$ with a theoretical and often unknown data generating process (DGP) $D(x)$. The parameters θ index the candidate distributions ~~used to describe the data~~, with their specific values typically encoding information about the data itself. In frequentist statistics these parameters θ are treated as fixed but unknown constants, however, when dealing with the Bayesian approach, the parameters are treated as random variables / distributions. In the most standard Bayes we have the posterior density

$$\pi_B(\theta | x_{1:n}) = \frac{\pi(\theta)L(x_{1:n} | \theta)}{m(x_{1:n})}.$$

Where $\pi(\theta)$ is our prior knowledge of the parameters θ , $L(x_{1:n} | \theta)$ is the likelihood function representing the probability of the observed data under those parameters, and $\pi_B(\theta | x_{1:n})$ is the pdf of the distribution $\theta | x_{1:n}$. The B here indicating that this is the standard Bayesian posterior. Here $m(x_{1:n}) = \int_\Theta L(x_{1:n} | \theta) d\pi(\theta)$ is the marginal normalising constant of the distribution, whose computation often makes the distribution intractable.

Bayesian inference is conceptually and practically appealing as, unlike frequentist statistical methods, we are able to allow our previous knowledge, specific domain expertise and experts in the subject matter to further inform our distribution. This injection of previous knowledge comes in the form of a carefully specified prior $\pi(\theta)$. Naturally, this comes with the potential risk of poor prior choice.

...

Further, Bayesian posteriors produce belief distributions (as opposed to point estimates, as an example, the direct outputs of MLE) over the parameters of interest $\theta \in \Theta$.

This carries two related benefits. First, prediction propagates parameter uncertainty rather than conditioning on a single value of θ ; where the conventional Bayesian predictive density for an unobserved x_{new} integrates the model over the posterior,

$$p(x_{\text{new}} | x_{1:n}) = \int_\Theta p(x_{\text{new}} | \theta) \pi_B(\theta | x_{1:n}) d\theta. \quad (1)$$

Second, any quantity of interest is recovered as a functional of the posterior, for instance the posterior mean $\hat{\theta}$ is recovered as $\int_\Theta \theta \pi_B(\theta | x_{1:n}) d\theta$. Because inference averages over θ instead of fixing it at a single best¹ value, the resulting predictions are less prone to the over-confidence of a plug-in point estimate.

¹Here *best* refers to the best available point estimate of θ , e.g. the MLE or MAP.

Moving to a more generalised Bayes, we can ~~change this to~~ **recast inference as** an optimisation-centric approach. Generalised Bayes replaces the likelihood with a loss function raised in an exponential.

$$\begin{aligned}\pi_w(\theta \mid x_{1:n}) &= \frac{\exp\left\{-w \sum_{i=1}^n \ell(\theta, x_i)\right\} \pi(\theta)}{\int_{\Theta} \exp\left\{-w \sum_{i=1}^n \ell(\theta, x_i)\right\} \pi(\theta) d\theta} \\ &\propto \exp\left\{-w \sum_{i=1}^n \ell(\theta, x_i)\right\} \pi(\theta)\end{aligned}$$

~~where once again there is a normaliser in the denominator that without we can find the distribution up to proportionality, as before, the denominator is the normalising constant; without it we characterise the distribution only up to proportionality.~~ There also exists a learning rate / loss scale $w \in \mathbb{R}_{>0}$ **(strictly positive; w rescales the total loss)**. This posterior is the Bissiri–Holmes–Walker (2016) Gibbs posterior.

It is important to note here that from this posterior we can recover our previous standard Bayes posterior π_B in the specific case with $\ell(\theta, x_i) = -\log p(x_i \mid \theta)$ and the learning rate $w = 1$. **The loss $-\log p(x_i \mid \theta)$ is the self-information (surprisal) of each observation, so this choice penalises the model exactly by how surprised it is by the data.** Following this we get

$$\begin{aligned}\pi_w(\theta \mid x_{1:n}) &\propto \exp\left\{-w \sum_{i=1}^n \ell(\theta, x_i)\right\} \pi(\theta) \\ &= \exp\left\{\sum_{i=1}^n \log p(x_i \mid \theta)\right\} \pi(\theta) \\ &= \left(\prod_{i=1}^n p(x_i \mid \theta)\right) \pi(\theta) \\ &= L(x_{1:n} \mid \theta) \pi(\theta)\end{aligned}$$

And further the normalising constant recovers the marginal likelihood in the same manner. Hence we recover $\pi_B(\theta \mid x_{1:n})$. **So standard Bayes is not a separate paradigm but a single specific instance of a more generalised approach where $\ell = \log$ loss, and $w = 1$.**

Aside: the learning rate w

Looking at the learning rate w here, we can see this as a weighting between the prior and the data, and a guard against model misspecification (assuming the prior is well specified). It acts as an effective-sample-size multiplier, effectively updating the prior with only wn data points. **The limiting behaviour makes this concrete:**

$$w \rightarrow 0^+ : \pi_w \rightarrow \pi \text{ (prior);} \quad w = 1, \log \text{ loss : } \pi_w = \pi_B; \quad w \rightarrow \infty : \pi_w \rightarrow \delta_{\hat{\theta}}, \quad \hat{\theta} = \arg \min_{\theta} \sum_i \ell(\theta, x_i).$$

The effect of this learning rate is exact and analytic and *does* affect the final distribution of the posterior. **Here w is part of the target distribution itself, not of any algorithm used to**

approximate it. This should be held separate from the step size of an optimiser, which (for a non-analytic solver) governs the trajectory toward a fixed target rather than the target. This distinction is flagged now because as later this property is altered. Once the posterior is itself defined as the solution of a particle optimisation (PVI), the data-weight and the optimiser's effect on the realised distribution no longer cleanly separate. Further discussed in Section ??.

Coherence and the loss-based view

Bissiri, Holmes & Walker do not posit the exponentiated-loss form; they derive it from a *coherence* requirement on how beliefs may be updated. Coherence demands that the update be self-consistent under partitioning of the data: updating on a first batch and then on a second must yield the same belief as updating on the pooled sample at once, so that the order and grouping of the data cannot change our conclusions. An update with (x_1, x_2) together or $\{(x_1), (x_2)\}$ separately yields the same result. This coherence property can be formally expressed as

$$\psi[l(\theta, x_2), \psi\{l(\theta, x_1), \pi(x)\}] = \psi\{l(\theta, x_2) + l(\theta, x_1), \pi(x)\},$$

where the function ψ is the update function such that in the standard case

$$\pi(\theta | x) \equiv \psi\{l(\theta; x), \pi(\theta)\}.$$

For an additive loss they show the unique update satisfying this is the Gibbs posterior above. This matters going forward because coherence is a property of the loss: the self-information (log) loss satisfies it, but as we adopt richer, predictive and discrepancy-based losses later, coherence is no longer automatic and must be checked rather than assumed.

Some of the motivation for the loss-based view is that one may often only be interested in a simple, low-dimensional statistic, such as a population mean or median. In standard Bayesian frameworks, estimating these simple statistics typically (As in pi_B) requires modelling the entire data distribution. Furthermore, some parameters of interest do not directly index any standard parametric family of density functions. (update / check mroe LATER) A loss can connect a parameter to data directly In our setting later on, we nonetheless retain the full posterior rather than a single functional, as ...

2 3

The optimisation-centric view and the Rule of Three

The generalised bayes posterior π_w has an equivalent expression focusing more on the optimisation side. In Knoblauch, Jewson & Damoulas (2022) it is shown that ~~and-setting-the~~

²absolute-error-loss inference can still succumb to misspecification, a robust loss limits but does not remove sensitivity to the DGP

³Link to PrOs, also the problem of w

multiplicative updating rule aside, the exact posterior can equally be obtained as the solution of an infinite-dimensional optimisation problem over the space $\mathcal{P}(\Theta)$ of all probability measures on Θ . references? maybe as called out: representation they trace to Csiszár (1975), Donsker & Varadhan (1975) and Zellner (1988):

$$\pi_B(\theta \mid x_{1:n}) = \arg \min_{q \in \mathcal{P}(\Theta)} \left\{ \mathbb{E}_q \left[\sum_{i=1}^n \ell(\theta, x_i) \right] + \text{KL}(q \parallel \pi) \right\}, \quad \ell(\theta, x_i) = -\log p(x_i \mid \theta).$$

Replacing the log loss with a general ℓ (and absorbing the loss scale w into ℓ) returns exactly the generalised Gibbs posterior of the previous section; Bissiri et al. (2016) show the minimiser exists and coincides with π_w ⁴ whenever the normaliser $Z = \int_{\Theta} \exp\{-\sum_i \ell(\theta, x_i)\} \pi(\theta) d\theta$ is finite. The objective can be interpreted as regularised empirical risk minimisation; it functions in two parts, minimising the expected loss over the data while the KL term pulls the solution toward the prior. These are both important points that will be further discussed later on. It should be noted that here it is the KL term which causes the optimisation problem to converge to a distribution, without it the problem collapses to a point mass $\delta_{\hat{\theta}}$ at the empirical risk minimiser $\hat{\theta} = \arg \min_{\theta} \sum_i \ell(\theta, x_i)$. This can be seen as we derive our previous bayes posteriors from this optimisation results.

Letting $\ell_n(\theta) = \sum_{i=1}^n \ell(\theta, x_i)$ denote the cumulative loss, the normalising constant Z and the normalised Gibbs posterior $\pi_w(\theta)$ are defined as

$$Z = \int_{\Theta} \pi(\theta) e^{-\ell_n(\theta)} d\theta \quad \text{and} \quad \pi_w(\theta) = \frac{\pi(\theta) e^{-\ell_n(\theta)}}{Z}.$$

The objective then collapses into a single Kullback-Leibler (KL) divergence,

$$\begin{aligned} \mathbb{E}_q \left[\sum_{i=1}^n \ell(\theta, x_i) \right] + \text{KL}(q \parallel \pi) &= \int_{\Theta} q(\theta) \ell_n(\theta) d\theta + \int_{\Theta} q(\theta) \log \frac{q(\theta)}{\pi(\theta)} d\theta \\ &= \int_{\Theta} q(\theta) \left[\log \frac{q(\theta)}{\pi(\theta)} + \ell_n(\theta) \right] d\theta \\ &= \int_{\Theta} q(\theta) \log \frac{q(\theta)}{\pi(\theta) e^{-\ell_n(\theta)}} d\theta \\ &= \int_{\Theta} q(\theta) \log \frac{q(\theta)}{Z \pi_w(\theta)} d\theta \\ &= \text{KL}(q \parallel \pi_w) - \log Z. \end{aligned}$$

Since $\log Z$ is independent of q and $\text{KL}(q \parallel \pi_w) \geq 0$ with equality if and only if $q = \pi_w$, the minimiser over $\mathcal{P}(\Theta)$ is exactly the Gibbs posterior,

$$\arg \min_{q \in \mathcal{P}(\Theta)} \left\{ \mathbb{E}_q \left[\sum_{i=1}^n \ell(\theta, x_i) \right] + \text{KL}(q \parallel \pi) \right\} = \pi_w(\theta) = \frac{\pi(\theta) \exp\{-\sum_{i=1}^n \ell(\theta, x_i)\}}{Z}.$$

⁴Knoblauch et al. (2022) do not notationally distinguish the standard and generalised Bayesian posteriors, writing q_B^* for both; here we keep them separate as π_B and π_w .

Specialising to the log loss $\ell(\theta, x_i) = -\log p(x_i | \theta)$ yields $e^{-\ell_n(\theta)} = \prod_{i=1}^n p(x_i | \theta) = L(x_{1:n} | \theta)$ alongside the marginal likelihood $Z = \int_{\Theta} \pi(\theta) L(x_{1:n} | \theta) d\theta = m(x_{1:n})$. Substituting these components gives

$$\pi_w(\theta) = \frac{\pi(\theta) L(x_{1:n} | \theta)}{m(x_{1:n})} = \pi_B(\theta | x_{1:n}),$$

which recovers the standard Bayesian posterior. **A quick proof of dericlt mass for without KL?**

Furthermore, we can see now that by using the Bayesian posterior, we are equivalently committing to this specific optimisation problem. As [paper ...] points out, similar to the generalised Bayes, using this standard posterior is only the right tool if minimising this objective actually reflects the overarching goal of the inference task. If the standard log loss does not align with those practical goals, this framework justifies replacing it with a custom loss function. (Knoblauch, Jewson & Damoulas, 2022).

This transformation to an optimisation problem is what we will focus on throughout this thesis, and from here on out.

Taking inference as an optimisation, the standard posterior is seen to fix three choices that need not be fixed. A loss ℓ , the divergence KL measuring departure from the prior, and the feasible set $\mathcal{P}(\Theta)$ optimised over. Knoblauch et al. free all three, defining the *Rule of Three* (RoT) posterior

$$\pi_{(\ell, D, \Pi)}^* := \arg \min_{q \in \Pi} \left\{ \mathbb{E}_q \left[\sum_{i=1}^n \ell(\theta, x_i) \right] + D(q \| \pi) \right\},$$

specified by a loss ℓ , a divergence D , and a feasible set $\Pi \subseteq \mathcal{P}(\Theta)$ ⁵. Standard Bayes is the choice $\pi_{(-\log p, \text{KL}, \mathcal{P}(\Theta))}^*$ and generalised Bayes is $\pi_{(\ell, \text{KL}, \mathcal{P}(\Theta))}^*$; the same template also recovers tempered and PAC-Bayesian posteriors. The three arguments ~~are not arbitrary:~~ they answer three assumptions that justify standard Bayes, ****but may have issues**** [⌋]. A well-specified prior (P), a well-specified likelihood (L), and unlimited computation (C). The divergence D controls how, and how strongly, the prior is enforced (P); the loss ℓ controls robustness to model misspecification (L); and the feasible set Π controls tractability (C).

An issue arises in this setup where, computationally, this optimisation becomes quite intractable over an infinite dimensional space; the third argument in our objective Π as the space of all possible probability measures $\mathcal{P}(\Theta)$, is infinite dimensional. Variational Inference helps this by restricting this third argument to a finite-dimensional family $\mathcal{Q} = \{q(\theta | \kappa) : \kappa \in K\}$, VI shrinks an impossible infinite-dimensional search down to a tractable finite one. Thus while VI is an approximation of the true posterior, it is still shown to be grounded by the same optimisation problem but constrained, yielding

$$\pi_{\text{VI}}^* = \pi_{(\ell, \text{KL}, \mathcal{Q})}^*.$$
⁶

⁵Notation: Knoblauch et al. write the RoT posterior as $P(\ell, D, \Pi)$. Here it is $\pi_{(\ell, D, \Pi)}^*$

⁶Knoblauch et al., 2022, Thm. 2

Standard VI is then not a separate approximation principle but the *best $\mathcal{Q}$ -constrained solution of the very same objective* (Knoblauch et al., 2022, Thm. 2). The tractable case in which $\Pi = \mathcal{Q}$ is a variational family and D may differ from KL is what they call *Generalised Variational Inference* (GVI). This shifts the role of $\mathcal{Q}$; rather than treating inference in $\mathcal{Q}$ as an approximation to π_B , we may take the objective itself as the design target and construct belief distributions in $\mathcal{Q}$ with the properties we want from the outset. **MoE-PVI is in this frame as a choice of feasible set Π (mixtures of experts) together with a [predictive / discrepancy-based] loss $\ell \dots$**

Variational and stochastic variational inference

Whether working within a standard or generalised framework, the resulting posterior typically remains intractable due to the normalising constant $p(x_{1:n})$, often referred to as the marginal likelihood or model evidence. This evidence represents the total probability of observing the data marginalised over all possible parameter configurations, calculated as $p(x_{1:n}) = \int_{\Theta} p(x_{1:n} | \theta) \pi(\theta) d\theta$. Because this high-dimensional integral rarely admits a closed-form solution, directly computing the posterior is usually impossible. Variational inference attempts to sidestep this integration bottleneck by restricting the search to a tractable family of distributions $\mathcal{Q}$ and selecting the member that is closest to the true posterior in Kullback-Leibler divergence. The ideal variational distribution is therefore

$$q^* = \arg \min_{q \in \mathcal{Q}} \text{KL}(q(\theta) \| \pi_B(\theta | x_{1:n})).$$

However, this divergence cannot be minimised directly because its evaluation depends on the very intractable evidence we are trying to avoid. Expanding the definition of the KL divergence reveals this dependency explicitly, yielding

$$\begin{aligned} \text{KL}(q(\theta) \| \pi_B(\theta | x_{1:n})) &= \int_{\Theta} q(\theta) \log \frac{q(\theta)}{\pi_B(\theta | x_{1:n})} d\theta \\ &= \int_{\Theta} q(\theta) \log \frac{q(\theta) p(x_{1:n})}{p(x_{1:n} | \theta) \pi(\theta)} d\theta \\ &= \int_{\Theta} q(\theta) [\log q(\theta) - \log \pi(\theta) - \log p(x_{1:n} | \theta)] d\theta + \log p(x_{1:n}) \\ &= \text{KL}(q(\theta) \| \pi(\theta)) - \mathbb{E}_q[\log p(x_{1:n} | \theta)] + \log p(x_{1:n}). \end{aligned}$$

By rearranging these terms, we can express the log evidence as the sum of two distinct components,

$$\log p(x_{1:n}) = (\mathbb{E}_q[\log p(x_{1:n} | \theta)] - \text{KL}(q(\theta) \| \pi(\theta))) + \text{KL}(q(\theta) \| \pi_B(\theta | x_{1:n})).$$

The bracketed term is known as the Evidence Lower Bound (ELBO). Because the KL divergence is strictly non-negative, the ELBO strictly lower-bounds the true log evidence. Furthermore, since the log evidence $p(x_{1:n})$ is a constant with respect to the variational

distribution q , minimising the KL divergence to the exact posterior is mathematically identical to maximising this lower bound, formally defined as

$$\text{ELBO}(q) = \mathbb{E}_q[\log p(x_{1:n} \mid \theta)] - \text{KL}(q(\theta) \parallel \pi(\theta)).$$

Classical coordinate-ascent variational inference optimises this ELBO with a full sweep over the dataset at each iteration, which quickly becomes computationally prohibitive for large datasets. To address this, Stochastic Variational Inference (Hoffman et al., 2013) applies stochastic optimisation by forming noisy, unbiased natural-gradient estimates of the ELBO from random data minibatches. By updating the parameters under a Robbins–Monro step-size schedule requiring $\sum_t \rho_t = \infty$ and $\sum_t \rho_t^2 < \infty$, the algorithm scales efficiently to massive datasets. While this natural gradient takes a remarkably simple form for conditionally conjugate exponential-family models, black-box variants (Ranganath et al., 2014) remove this conjugacy requirement entirely by relying on score-function gradient estimators. However, the defining limitation for all of these approaches is that they fix a parametric family $\mathcal{Q}$ in advance. The quality of the final approximation can never exceed the expressiveness of this chosen family, which is precisely the rigid restriction that the particle-based methods discussed in Section 3 are designed to relax.

2 2:

New statistical principle that combines desirable aspects of both parameter inference and density estimation. Uncertainty as a consequence of predictive ability. The discrepancy from (1).

PrO posteriors converge at $n^{-\frac{1}{2}}$ while conventional only converge at this rate if the model can recover the true DGP.

$$\pi_w = \arg \min_{q \in \mathcal{P}(\Theta)} \left\{ \frac{\lambda_n}{n} \sum_{i=1}^n \int_{\Theta} S(P_{\theta}, x_i) dq(\theta) + \text{KL}(q \parallel \pi) \right\}$$

$$\pi_{\text{PrO}} = \arg \min_{q \in \mathcal{P}(\Theta)} \left\{ \frac{\lambda_n}{n} \sum_{i=1}^n S(P_q, x_i) + \text{KL}(q \parallel \pi) \right\}, \quad P_q := \int_{\Theta} P_{\theta} dq(\theta) \quad (2)$$

$$d\pi_{\text{PrO}}(\theta) \propto \pi(\theta) \exp \left\{ -\frac{\lambda_n}{n} \sum_{i=1}^n \frac{\delta S(P_q, x_i)}{\delta q} \Big|_{q=\pi_{\text{PrO}}}(\theta) \right\} \quad (3)$$

Gibbs minimises the average of scores; PrO minimises the score of the average.

3 3:

At the end of Section 1 we saw a structural limitation with standard Variational and stochastic variational inference. That is, to make our approximation tractable we fix a parametric family

$\mathcal{Q}$ in advance. However, by setting an approximating distribution we limit how rich our expression can be to the maximum the selected family permits; under model misspecification, where the best description of the data may be multimodal or heavy-tailed, this restriction is what degrades the approximation.

With particle-based methods (or particle approximations), we remove this constraint. Instead of parameterising q (where q is our variational approximation of the true target distribution π_w as noted in section 1 (elbo)) and optimising q 's parameters, we use a cloud of M particles $\{\theta_1, \dots, \theta_M\}$ with the empirical measure $q \approx \frac{1}{M} \sum_j \delta_{\theta_j}$ and transport the particles directly. By transitioning to particle-based methods, we redefine the feasible set Π within the Rule of Three framework, replacing the rigid parametric $\mathcal{Q}$ with the infinitely flexible space of empirical measures.

Two complementary lineages dominate the literature: the deterministic Stein flow of Liu and Wang (2016), and the Wasserstein-gradient-flow view of Wild et al. (2023), which unifies deep ensembles, Langevin dynamics and variational inference under a single functional. This section reviews both, and reconstructs the second on the toy problems of Wild et al. to expose the machinery our own method inherits.

3.1 Stein variational gradient descent (SVGD)

4 Lit review

1. Foundations of Belief and Updating & Variational Inference

- A general framework for updating belief distributions (Bissiri et al.)
- Stochastic Variational Inference
- An optimisation-centric view on Bayes' rule: Reviewing and generalising variational inference (Knoblauch et al.)

2. The Shift to Predictively Oriented Posteriors

- Predictive Variational Inference: Learning the predictively optimal posterior distribution
- Predictive Variational Inference for flexible regression models
- Predictively Oriented Posteriors (McLatchie et al.)

3. Particle-Based Inference, MMD, and Misspecification

- Stein Variational Gradient Descent (Liu & Wang) (Very quick mention) and time complexity things
- A Rigorous Link between Deep Ensembles and (Variational) Bayesian Methods

- Prediction-Centric Uncertainty Quantification via MMD (Shen et al.), Wild et al. 2023
- maybe: (Detecting model misspecification in Bayesian inverse problems via variational gradient descent)

4. **Variational Inference with Normalising Flows?? Adaptive Heterogeneous Mixtures of Normalising Flows for Robust Variational Inference?? And more MOE papers...**